Abhishek Lakhani

Applied AI Engineer – NLP & LLM Systems | M.Sc. Student

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Work authorisation: no visa sponsorship required | English (fluent), German (B1)

PROFILE

M.Sc. student in Automotive Software Engineering at TU Chemnitz who builds and measures NLP and LLM systems in Python. I have built and published tested projects covering hybrid RAG and semantic search (FastAPI, Qdrant, nDCG@10 0.737), LLM structured extraction with Pydantic validation, a fine-tuned German DistilBERT classifier (0.897 macro-F1) and its ONNX int8 deployment (1.46x throughput, 4x smaller). Practical experience from a machine learning internship (federated learning, TensorFlow) and two software internships. Looking for working-student, thesis or junior roles in Applied AI and ML engineering.

SKILLS

Programming: Python, SQL, JavaScript, TypeScript, HTML/CSS

LLM & NLP: LLMs, Prompt Engineering, Few-Shot Prompting, Structured Outputs, RAG, LangChain, Hugging Face Transformers, Fine-Tuning (DistilBERT), Sentence Embeddings, Text Classification, Information Retrieval, LLM APIs (Anthropic, Ollama)

Search & Vector DB: Qdrant, FAISS, BM25, TF-IDF, Hybrid Search (RRF), Cross-Encoder Reranking

Machine Learning: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Model Evaluation (nDCG, MRR, Recall@k, F1, Significance Tests)

MLOps & Backend: FastAPI, REST APIs, Pydantic, Docker, Docker Compose, ONNX Runtime, Quantization, pytest, GitHub Actions (CI), SQLAlchemy, PostgreSQL, Streamlit

Tools & Methods: Git, GitHub, Linux, n8n, CVAT, Agile/Scrum, Code Reviews

PROFESSIONAL EXPERIENCE

Machine Learning Intern – Technische Universität Chemnitz, Chemnitz, Germany | 06/2025 – 01/2026

- Designed a privacy-focused image classification pipeline in TensorFlow and ran federated learning experiments across decentralised Solid Pod data sources.
- Built modular Python components (structured logging, custom exceptions, CSV-based result tracking) so that experiments could be reproduced and compared.

Software Development Intern – Wokontech IT Solution, Surat, India | 10/2021 – 03/2022

- Built and trained object detection pipelines on a dataset of 5,000+ images and improved model robustness through data curation.
- Evaluated models with standard detection metrics and ran the annotation workflow in CVAT.

Backend Development Intern – White Orange Software, Surat, India | 12/2020 – 05/2021

- Built and maintained backend features in Core PHP and tuned MySQL queries for performance.
- Took part in code reviews and wrote technical documentation in a cross-functional team.

PROJECTS

RAG Document Q&A Service | Python, FastAPI, BM25, BGE Embeddings (ONNX), OCR, SSE, Docker
github.com/abhisheklakhani-it/rag-docqa

- Built a retrieval-augmented generation (RAG) service that answers questions on uploaded PDFs, including scanned PDFs via OCR, with cited answers streamed over Server-Sent Events.
- Hybrid BM25 + dense retrieval with reciprocal rank fusion raised nDCG@10 from 0.567 (TF-IDF) to 0.728 on BEIR SciFact; a calibrated similarity floor rejects 87% of off-topic questions. 73 automated tests, multi-stage Docker image.

Semantic Search API | FastAPI (async), Qdrant, Hybrid Dense + Sparse Search, Docker Compose
github.com/abhisheklakhani-it/semantic-search-api

- Production-style REST API with hybrid search inside Qdrant (nDCG@10 0.737 vs 0.715 dense-only), metadata filters, pagination, health checks and JSON logs with request IDs.
- Benchmarked a cross-encoder reranker and kept it off by default because it lowered quality on SciFact and cost 25–90x the latency. 33 automated tests.

ONNX Inference Optimisation | ONNX Runtime, Graph Fusion, int8 Quantization, FastAPI, Docker
github.com/abhisheklakhani-it/onnx-inference-optimization

- Exported a fine-tuned German DistilBERT to ONNX with graph fusion and int8 dynamic quantization: 1.46x PyTorch throughput, 4x smaller model (68 MB) for -0.3 points accuracy, served from a 147 MB Docker image.

German Text Classifier | Hugging Face Transformers, PyTorch, Qdrant, multilingual-e5
github.com/abhisheklakhani-it/german-text-classifier

- Fine-tuned distilbert-base-german-cased on 10kGNAD news (9 topics): 0.897 macro-F1 versus 0.845 for an embeddings + Qdrant kNN baseline; McNemar test $p < 0.001$.

LLM Structured Extractor | LLM Structured Outputs, Pydantic v2, FastAPI
github.com/abhisheklakhani-it/llm-structured-extractor

- Extracts validated JSON (title, company, skills, salary, remote policy) from English and German job postings with schema validation and retry on error; error-driven prompt changes raised all-fields-correct on a held-out set from 0.40 to 0.67 (mean of 3 runs). 43 automated tests.

Retrieval Evaluation Benchmark | scikit-learn, FAISS, sentence-transformers, pandas
github.com/abhisheklakhani-it/retrieval-eval-bench

- Compared TF-IDF, BM25, dense and hybrid retrieval on BEIR SciFact with a train/test split and paired significance tests; hybrid search reached nDCG@10 0.734 ($p = 0.013$ vs best single method).

Test Bench Commissioning Web App | FastAPI, SQLAlchemy, PostgreSQL, Docker, GitHub Actions, n8n
github.com/abhisheklakhani-it/n8n-test-bench-commissioning-tracker

- Role-based web app that digitalises test bench commissioning (Prüfstand-Inbetriebnahme) with priority notifications, a tablet shop-floor mode and an audit log; 137 tests, CI with dependency audit, live demo. Grew out of an n8n workflow prototype.

LinkedIn Post Generator | LangChain, Few-Shot Prompting, Pydantic, Streamlit
github.com/abhisheklakhani-it/linkedin-post-generator

- LangChain app that writes English/German posts as validated structured output and flags any number the user never supplied, to catch invented facts. 34 automated tests.

EDUCATION

M.Sc. Automotive Software Engineering – Technische Universität Chemnitz, Chemnitz, Germany | 04/2022 – present

- Master's thesis (in preparation): Real-time LiDAR health monitoring system with GUI for railway applications.

B.E. Computer Science and Engineering – Gujarat Technological University (Government Engineering College Dahod), Gujarat, India | 08/2017 – 07/2021

LANGUAGES

English – fluent | German – B1 (improving)

VOLUNTEERING

- Erasmus+ Youth Exchange: international project work on social topics, including a project in France on gender equality.
- Volunteer at a public cycling event in Chemnitz (participant support and route guidance).